

Invariant Graph Subspaces Missing from the Claimed Lattices of Powers of Unicellular Weighted Shifts

Candidate counterexample packet for arXiv:2205.00252

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Status: candidate counterexample likely valid. The construction gives a negative answer to the coordinate-form persistence question in Remark 4.2. More strongly, it supplies continuum many counterexamples to Theorems 3.9–3.11 under the papers' own weight hypotheses.

1 Source and exact question

Lata, Pokhriyal, and Singh study powers of unicellular weighted shifts in [1]. In Remark 4.2 (arXiv PDF page 20; journal page 257), they ask whether the lattice descriptions in Theorems 3.9–3.11 remain the same for other sufficient classes of unicellular weights.

Remark 4.2. *As demonstrated in Sections 2 and 3, weights have no bearing on our characterization of finite-dimensional invariant subspaces of the operators T^{*2} and T^{*3} , but they are crucial for the infinite-dimensional cases. Interestingly, the lattice structure of the invariant subspaces of T^{*2} and T^{*3} is the same (Theorem 3.9, 3.10 and 3.11) for two independent classes of weights that we have considered. There are numerous more sufficient conditions on the weights $\{w_n\}_{n \in \mathbb{N}}$ for the weighted shift to be unicellular, see [5, 6, 8, 9, 10, 16, 17, 18, 22]. So the first critical question is whether the lattice structure of the invariant subspaces of T^{*2} and T^{*3} remains the same for these other weights also? And, if the answer to this question is affirmative, then developing a single strategy that works for all these weight classes will be an extremely interesting problem to work on.*

In particular, Theorem 3.10 claims that, under the monotone square-summable weight condition (3.4), every proper infinite-dimensional T^{*2} -invariant subspace has one of a countable collection of coordinate forms. Theorem 3.11 gives an analogous coordinate description for T^{*3} .

2 Definitions

Let $\mathcal{H} = \ell^2(\mathbb{N}_0)$ with orthonormal basis $(e_n)_{n \geq 0}$. For a bounded sequence of positive weights (w_n) , let $B = T^*$ be the backward weighted shift

$$Be_0 = 0, \quad Be_n = w_{n-1}e_{n-1} \quad (n \geq 1).$$

For an integer $m \geq 2$ and $0 \leq r < m$, define the residue subspace

$$\mathcal{H}_r^{(m)} = \overline{\text{span}}\{e_{mk+r} : k \geq 0\}.$$

Then $B^m \mathcal{H}_r^{(m)} \subseteq \mathcal{H}_r^{(m)}$, while $B \mathcal{H}_r^{(m)} \subseteq \mathcal{H}_{r-1}^{(m)}$ when $1 \leq r < m$.

3 Idea of the proof

The power B^m preserves each residue class, but it also commutes with B . Consequently the graph of the restriction $B : \mathcal{H}_r^{(m)} \rightarrow \mathcal{H}_{r-1}^{(m)}$ is B^m -invariant. The two residue spaces are orthogonal, so this graph is automatically closed and cannot be a coordinate subspace. Varying its scalar slope produces continuum many such graphs. This elementary family is absent from the source paper's lists.

4 Universal invariant-graph construction

Theorem 1. *Let B be any backward weighted shift with positive bounded weights. Fix $m \geq 2$, $1 \leq r < m$, and $\lambda \in \mathbb{C}$. Then*

$$\mathcal{G}_{\lambda,m,r} := (I + \lambda B)\mathcal{H}_r^{(m)}$$

is a closed infinite-dimensional B^m -invariant subspace. If $\lambda \neq 0$, it is proper and

$$\mathcal{G}_{\lambda,m,r} \cap \mathcal{H}_r^{(m)} = \mathcal{G}_{\lambda,m,r} \cap \mathcal{H}_{r-1}^{(m)} = \{0\}.$$

For fixed m, r , distinct values of λ give distinct subspaces.

Proof. For $x \in \mathcal{H}_r^{(m)}$, the vectors x and Bx belong to orthogonal residue spaces. Hence

$$\|(I + \lambda B)x\|^2 = \|x\|^2 + |\lambda|^2 \|Bx\|^2 \geq \|x\|^2.$$

Thus $I + \lambda B$ is bounded below on $\mathcal{H}_r^{(m)}$, so its range there is closed and infinite-dimensional. Since B commutes with B^m and $B^m\mathcal{H}_r^{(m)} \subseteq \mathcal{H}_r^{(m)}$,

$$B^m(I + \lambda B)x = (I + \lambda B)B^m x \in \mathcal{G}_{\lambda,m,r}.$$

Assume $\lambda \neq 0$. The restriction of B to $\mathcal{H}_r^{(m)}$ is injective because $r \geq 1$ and all weights are positive. If $x + \lambda Bx$ lies in $\mathcal{H}_r^{(m)}$, its orthogonal $\mathcal{H}_{r-1}^{(m)}$ component gives $Bx = 0$, hence $x = 0$. If it lies in $\mathcal{H}_{r-1}^{(m)}$, its $\mathcal{H}_r^{(m)}$ component gives $x = 0$. The two stated intersections are therefore trivial, and in particular the graph is proper.

Finally, if $\mathcal{G}_{\lambda,m,r} = \mathcal{G}_{\mu,m,r}$, comparison of the $\mathcal{H}_r^{(m)}$ components shows that a vector x must have the same graph representative on both sides. Thus $(\lambda - \mu)Bx = 0$ for every x , and injectivity gives $\lambda = \mu$. $\square$

5 Counterexamples to the source statements

Take the Donoghue weights

$$w_n = 2^{-n}.$$

They are positive, monotonically decreasing, and square summable, so they satisfy Condition (3.4) of [1]. They also satisfy Condition (3.1): for a product of length $L \geq 1$, the series occurring there is $\sum_{k \geq 0} 2^{-2kL} \leq (1 - 2^{-2})^{-1}$. Set

$$\mathcal{H}_{\text{odd}} = \overline{\text{span}}\{e_{2k+1} : k \geq 0\}, \quad \mathcal{G} = (I + B)\mathcal{H}_{\text{odd}}.$$

By Theorem 1, $\mathcal{G}$ is closed, proper, infinite-dimensional, and B^2 -invariant. It meets both the even and odd coordinate subspaces only at zero. It is therefore neither coordinate residue space in Theorems 3.9(i) and 3.10(i). Every space in parts (ii) of those theorems contains e_0 , whereas $\mathcal{G}$ does not. Thus $\mathcal{G}$ is absent from the claimed exhaustive lists. Varying the nonzero scalar λ gives continuum many missing spaces, while each displayed list is countable.

The same argument with $m = 3$ and $r = 1$ contradicts Theorem 3.11: the graph has zero intersection with each residue summand occurring in it, whereas every nonzero coordinate direct sum in that theorem contains a pure residue vector. The forms in Theorem 3.11(iii) additionally contain e_0 .

Consequently the premise behind Remark 4.2 is already false under one of the paper’s two original weight classes. In particular, its coordinate-form persistence question has a negative answer.

6 The lattices genuinely depend on the weights

The preceding objection is not merely that the common lists are incomplete. As literal collections of subspaces, the lattices can differ even among geometric weights satisfying Condition (3.4).

For $0 < q < 1$, let

$$B_q e_n = q^{n-1} e_{n-1} \quad (n \geq 1), \quad \mathcal{G}_q = (I + B_q) \mathcal{H}_{\text{odd}}.$$

Then $\mathcal{G}_q \in \text{Lat}(B_q^2)$. Let $p \in (0, 1)$ and consider $g = e_3 + q^2 e_2 \in \mathcal{G}_q$. Direct calculation gives

$$B_p^2 g = p^3 e_1 + q^2 p e_0.$$

If this vector belonged to $\mathcal{G}_q$, its odd component would force its graph preimage to be $p^3 e_1$, whose even component is $B_q(p^3 e_1) = p^3 e_0$. Thus membership would require $q^2 p = p^3$, or $q = p$. Taking $q = 1/2$ and $p = 1/3$ proves

$$\mathcal{G}_{1/2} \in \text{Lat}(B_{1/2}^2) \quad \text{but} \quad \mathcal{G}_{1/2} \notin \text{Lat}(B_{1/3}^2).$$

Both weight sequences are within Condition (3.4).

7 Verification and limitations

The script `code/verify_counterexample.py` uses exact rational matrices. It checks the graph construction for all residues and powers $2 \leq m \leq 6$, distinct slopes, the square counterexample, and the $q = 1/2$ versus $p = 1/3$ failure. Its output is:

```
all exact finite-section checks passed
checked graph invariance for powers 2 through 6
verified the q=1/2 graph is not invariant for the p=1/3 square
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The computation is a guard against indexing errors; the proof is the infinite-dimensional argument above.

A bounded novelty search on 11 August 2026 used the exact title, arXiv id, authors, theorem number, “lattice structure”, and combinations of “weighted shift”, “power”, and “invariant graph subspace”. It found the arXiv version and the 2023 journal publication, but no correction, erratum, or later source explicitly recording this counterexample. This is not a certification of novelty.

Human-review recommendation. Check the graph invariance identity and compare the two trivial-intersection properties directly with Theorems 3.10 and 3.11. Because the conclusion contradicts published characterization statements, external use should wait for independent expert review.

References

- [1] S. Lata, S. Pokhriyal, and D. Singh, *Invariant subspaces of powers of some unicellular operators*, New York J. Math. **29** (2023), 231–260; [arXiv:2205.00252](https://arxiv.org/abs/2205.00252).